

Capstone hypothesis and baseline

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can write a testable capstone hypothesis and a credible baseline.

I can specify data roles and a selection rule before inspecting results.

I can calculate baseline MSE and distinguish a success criterion from a result.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Write a testable capstone hypothesis and a credible baseline

Specify data roles and a selection rule before inspecting results

Calculate baseline MSE and distinguish a success criterion from a result

Retrieve what you already know

Why must a checkpoint chosen using validation data still face an untouched final test?

My starting explanation

This week's end goal

Explain capstone hypothesis and baseline with a traceable toy-network case and a bounded conclusion.

2. Notice the evidence

A capstone needs a question that evidence can answer. "Build a better network" leaves better undefined. Here the question is: does two-update full-batch fitting of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight baseline on a constructed positive-input task? Predict $\max(0, a \cdot x)$, with no bias. The baseline has $a = 1$. The trained candidate starts at $a = 1$, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

The hypothesis is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests training and record keeping, not whether neural networks are preferable to every other model. A stronger comparison may also include an ordinary fitted line.

List the given facts and one fact that is not supplied.

3. See how it works

Role and ID · x · y

Role and ID: Training T1

x: 1

y: 2

Role and ID: Training T2

x: 2

y: 4

Role and ID: Validation V1

x: 3

y: 6

Role and ID: Validation V2

x: 4

y: 8

Role and ID: Final test

x: Sealed

y: Sealed

Training rows determine updates. Validation rows compare the prespecified candidate with the baseline. The final-test inputs and targets remain sealed until the procedure and reporting plan are frozen. MSE = sum of squared prediction errors divided by the number of evaluated rows. Baseline predictions on V1 and V2 are 3 and 4. Errors are -3 and -4, squared errors 9 and 16, and validation MSE is $25/2 = 12.5$. There are two validation cases.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: T1 and T2

Procedure: Fit two updates

Permitted conclusion: Training result only

Plan or input: V1 and V2

Procedure: Compare with $a = 1$

Permitted conclusion: Validation comparison

Plan or input: Sealed final rows

Procedure: Evaluate frozen procedure

Permitted conclusion: Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Compute baseline predictions, errors, squared errors and MSE on V1 and V2. State the denominator.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Give only B the eight evaluation labels during training.

Prediction: The proposed comparison becomes contaminated rather than a valid held-out test.

Reason: One version has gained privileged information from the reserved evaluation set.

5. Words that unlock the idea

Word · Meaning

Word: Hypothesis

Meaning: A testable expectation with a stated comparison and outcome

Word: Baseline

Meaning: A specified reference procedure evaluated on the same cases

Word: Success criterion

Meaning: A rule written before evaluation for judging the measured result

Word: Reserved test

Meaning: Cases kept out of fitting, tuning and selection

Use two terms to explain a step in this week's example.

6. Follow a worked example

The validation criterion is candidate MSE strictly below baseline MSE on V1 and V2. Equality does not pass. The final criterion is lower MSE than the same baseline on the sealed final rows. Report both scores and every raw prediction whether or not the criterion passes. Record the rule, training rows, update count, learning rate, loss, data versions and code version. A criterion specifies how to judge a future result; it is not evidence that the result already happened.

Keep validation and final criteria separate. Do not silently add epochs or change preprocessing after seeing a weak result. A changed plan is a new version and needs an honest account of what evidence informed it.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone hypothesis and baseline

A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

The facts we will use

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the prediction

B should have more correct labels than A on the same eight patterns.



2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.



3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Write the prediction

What happens and why: B should have more correct labels than A on the same eight patterns.

Step: 2. Declare the baseline

What happens and why: A is a documented corner-only network, not a deliberately broken version.

Step: 3. Reserve evaluation

What happens and why: Keep the eight patterns out of model selection and tuning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A capstone hypothesis links a specific architectural change to a measured outcome. A baseline anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

What this example does not establish

Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

7. Find and repair a mistake

Supported example: a separate baseline predicts 2 and 4 for validation targets 5 and 7. Its errors are -3 and -3 and MSE is 9. A proposed plan says to train one fixed candidate and report both MSE values on those same two rows. It supplies no candidate predictions, so no improvement can yet be claimed.

Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.

Locate the earliest unsupported step

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

State one repair and the checks you would rerun.

8. Practise together

State the prediction rule, baseline, candidate learning rate and update count, comparison data and metric. Write a testable hypothesis.

Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.

Explain your prediction to a partner before comparing results.

Practise with less help: Calculate the reference score

Use workbook W2 for the connected example “Capstone hypothesis and baseline”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone hypothesis and baseline. Proposed explanation: “A hypothesis is a result”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Name the intended change, the baseline and the measured outcome. Why must both versions avoid training on the eight reserved labels?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Compute baseline predictions, errors, squared errors and MSE on V1 and V2. State the denominator.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

State the prediction rule, baseline, candidate learning rate and update count, comparison data and metric. Write a testable hypothesis.

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent evidence.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

For Image-pattern Network Investigation, use this week’s independent task as a rehearsal for the final evidence pack: A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning

rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet? Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

12. Show what you know

Before running, exchange a plan with a partner. The partner should be able to identify what is changed, what stays fixed, what metric is computed, which rows select the procedure and when the final test may be opened. A file name alone does not protect a split; record who can inspect it and how related examples were separated.

Exit: someone opens the final targets to choose between learning rates, then calls the winning score independent final evidence. Explain the change of role and the next honest step.

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent evidence.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: T1 and T2

Procedure: Fit two updates

Permitted conclusion: Training result only

Plan or input: V1 and V2

Procedure: Compare with $a = 1$

Permitted conclusion: Validation comparison

Plan or input: Sealed final rows

Procedure: Evaluate frozen procedure

Permitted conclusion: Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain

every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Plan a new case independently

A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.